CS 556: Mathematical Foundations of Machine Learning
Assignment 1

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Problem 1

Solve the following system of linear equations by elimination and back substitution:

$$\begin{aligned} x + 2y + 3z &= 0 \\ -x + y - 2z &= -3 \\ 2x + y + z &= 3 \end{aligned}$$

$$(1) + (2) \quad \Longrightarrow \quad 3y + z = -3$$

$$(3) - 2(1) \quad \Longrightarrow \quad -3y - 5z = 3$$

$$\begin{aligned} (3y + z) + (-3y - 5z) &= -3 + 3 \\ -4z &= 0 \\ \Longrightarrow \quad z &= 0 \end{aligned}$$

Back substitution:

$$3y + z = -3 \quad \xrightarrow{z=0} \quad 3y = -3 \quad \Longrightarrow \quad y = -1$$

$$x + 2y + 3z = 0 \quad \xrightarrow{y=-1,\, z=0} \quad x - 2 = 0 \quad \Longrightarrow \quad x = 2$$

$$\boxed{(x, y, z) = (2, -1, 0)}$$

Problem 2

Let A be any $n \times n$ matrix.

(a)

If A is invertible, show that A^2 is invertible.

Since A is invertible,

$$AA^{-1} = A^{-1}A = I.$$

Then

$$\begin{aligned} A^2(A^{-1})^2 &= (AA)(A^{-1}A^{-1}) \\ &= A(AA^{-1})A^{-1} \\ &= AIA^{-1} \\ &= AA^{-1} \\ &= I. \end{aligned}$$

Likewise,

$$(A^{-1})^2A^2 = I.$$

Therefore,

$$\boxed{(A^2)^{-1} = (A^{-1})^2}$$

and hence

$$\boxed{A^2 \text{ is invertible.}}$$

(b)

Suppose A^2 is invertible. Can A be singular? If yes, give such an example. If not, explain why A is non-singular.

$$A^2 \text{ invertible} \quad \Longrightarrow \quad \det(A^2) \neq 0$$

$$\det(A^2) = \det(A)\det(A) = (\det A)^2$$

Therefore,

$$(\det A)^2 \neq 0 \quad \Longrightarrow \quad \det(A) \neq 0$$

$$\det(A) \neq 0 \quad \Longrightarrow \quad A \text{ is invertible}$$

Thus,

$$\boxed{A \text{ cannot be singular.}}$$

Problem 3

Express

$$x = \begin{bmatrix} 1 \\ -2 \\ 7 \end{bmatrix}$$

as a linear combination of

$$a = \begin{bmatrix} 2 \\ 4 \\ 4 \end{bmatrix}, \qquad b = \begin{bmatrix} -3 \\ -6 \\ 3 \end{bmatrix}, \qquad c = \begin{bmatrix} 4 \\ 10 \\ 10 \end{bmatrix}.$$

Let

$$x = \alpha a + \beta b + \gamma c.$$

Then

$$\begin{bmatrix} 1 \\ -2 \\ 7 \end{bmatrix} = \alpha \begin{bmatrix} 2 \\ 4 \\ 4 \end{bmatrix} + \beta \begin{bmatrix} -3 \\ -6 \\ 3 \end{bmatrix} + \gamma \begin{bmatrix} 4 \\ 10 \\ 10 \end{bmatrix}.$$

Thus,

$$\begin{aligned} 2\alpha - 3\beta + 4\gamma &= 1 \\ 4\alpha - 6\beta + 10\gamma &= -2 \\ 4\alpha + 3\beta + 10\gamma &= 7 \end{aligned}$$

Eliminate α:

$$(2) - 2(1) \quad \Longrightarrow \quad 2\gamma = -4 \quad \Longrightarrow \quad \gamma = -2$$

$$(3) - 2(1) \quad \Longrightarrow \quad 9\beta + 2\gamma = 5$$

$$9\beta + 2(-2) = 5 \quad \Longrightarrow \quad 9\beta = 9 \quad \Longrightarrow \quad \beta = 1$$

Back substitution into (1):

$$2\alpha - 3(1) + 4(-2) = 1$$

$$2\alpha - 11 = 1 \quad \Longrightarrow \quad 2\alpha = 12 \quad \Longrightarrow \quad \alpha = 6$$

Therefore,

$$\boxed{x = 6a + b - 2c}$$

Problem 4

If $\|v\| = 5$ and $\|w\| = 3$, what are the smallest and largest values of $\|v - w\|$? What are the smallest and largest values of $v \cdot w$?

By the triangle inequality,

$$|\|v\| - \|w\|| \leq \|v - w\| \leq \|v\| + \|w\|$$

$$|5 - 3| \leq \|v - w\| \leq 5 + 3$$

$$\boxed{2 \le \|v - w\| \le 8}$$

Therefore,

$$\boxed{\min \|v - w\| = 2, \qquad \max \|v - w\| = 8}$$

For the dot product,

$$v \cdot w = \|v\|\|w\| \cos\theta$$

$$v \cdot w = (5)(3)\cos\theta = 15\cos\theta$$

Since

$$-1 \le \cos\theta \le 1,$$

$$-15 \le 15\cos\theta \le 15$$

so

$$\boxed{\min(v \cdot w) = -15, \qquad \max(v \cdot w) = 15}$$

Problem 5

Find the determinant of

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 3 & 2 \\ 1 & -1 & -2 \end{bmatrix}.$$

Expanding along the first row,

$$\begin{aligned} \det(A) &= 1\begin{vmatrix} 3 & 2 \\ -1 & -2 \end{vmatrix} - 0\begin{vmatrix} 0 & 2 \\ 1 & -2 \end{vmatrix} + 1\begin{vmatrix} 0 & 3 \\ 1 & -1 \end{vmatrix} \\ &= 1\,[3(-2) - 2(-1)] + 1\,[0(-1) - 3(1)] \\ &= (-6 + 2) + (0 - 3) \\ &= -4 - 3 \\ &= -7. \end{aligned}$$

$$\boxed{\det(A) = -7}$$

Problem 6

Use Gauss–Jordan elimination to find A^{-1} if

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & -1 & -1 \\ 1 & 0 & 1 \end{bmatrix}.$$

Start with $[A \mid I]$:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & -1 & -1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R_3 \leftarrow R_3 - R_1} \left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & -1 & -1 & 0 & 1 & 0 \\ 0 & 0 & -1 & -1 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R_3 \leftarrow -R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & -1 & -1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & -1 \end{array}\right]$$

$$\xrightarrow[R_2 \leftarrow R_2 + R_3]{R_1 \leftarrow R_1 - 2R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & 0 & 2 \\ 0 & -1 & 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 1 & 0 & -1 \end{array}\right]$$

$$\xrightarrow{R_2 \leftarrow -R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & 0 & 2 \\ 0 & 1 & 0 & -1 & -1 & 1 \\ 0 & 0 & 1 & 1 & 0 & -1 \end{array}\right]$$

Thus,

$$\left[I \mid A^{-1}\right] = \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & 0 & 2 \\ 0 & 1 & 0 & -1 & -1 & 1 \\ 0 & 0 & 1 & 1 & 0 & -1 \end{array}\right],$$

so

$$\boxed{A^{-1} = \begin{bmatrix} -1 & 0 & 2 \\ -1 & -1 & 1 \\ 1 & 0 & -1 \end{bmatrix}}$$